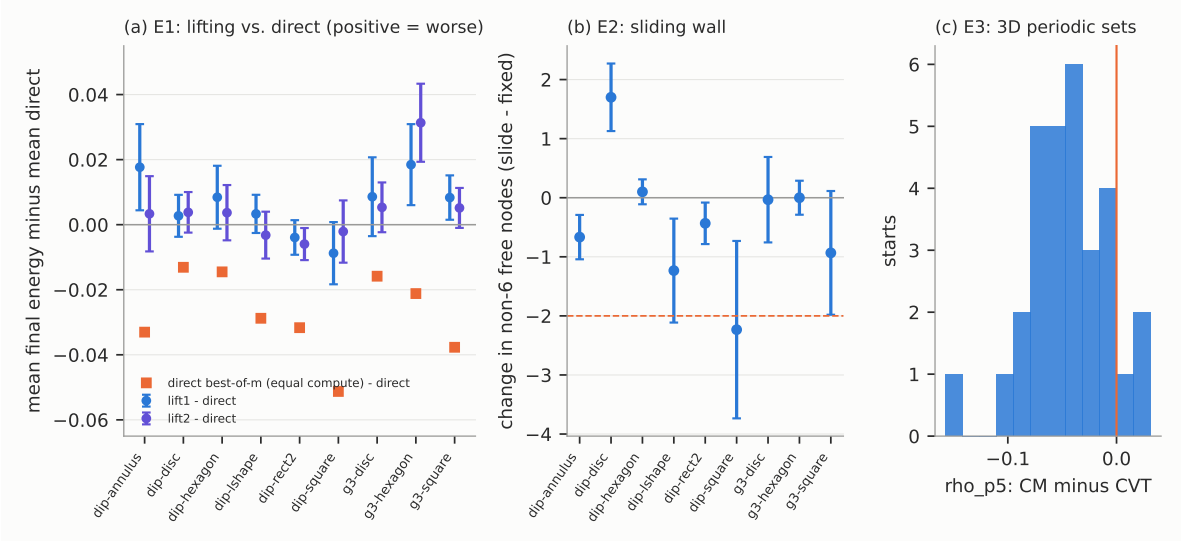


(a) Mean final energy of the lifted runs minus the direct mean (valid runs, ±1 s.e. of the difference); orange: best of m direct restarts (m = evaluation ratio of lift1) minus the direct mean. (b) Change of the number of non-6-coordinated free nodes with sliding walls, ±1 s.e.; dashed: registered threshold −2. (c) Paired difference of the 5th-percentile tetrahedron radius ratio, CM minus CVT, 30 starts.

Gated exciton fluids IV:

three heuristics tested and not confirmed: dimensional lifting, sliding walls, and completely monotone relaxation as a 3D mesh generator

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Abstract

After papers I–III [1, 2, 3], three ideas that had been suggested for the classical node-arrangement problem were pre-registered and tested with a plain numpy L-BFGS engine on the container geometries of the Fenris study: **(E1)** lifting the particles to $2 + k$ dimensions and annealing the extra coordinates away, as an optimisation aid; **(E2)** letting the wall nodes slide along the wall; **(E3)** using a completely monotone repulsion in a periodic cube as a 3D mesh generator, against Lloyd/centroidal Voronoi relaxation. Results: E1 shows no benefit at equal compute in any of the nine shape–kernel cases (30 starts each); E2 reduces the number of non-sixfold-coordinated nodes by 2 or more in one of nine cases and increases it significantly in one; E3 is slightly *worse* than CVT in the 5th-percentile tetrahedron radius ratio (30 paired starts). All three registered gates are not met. The runs are small ($N_{\text{free}} \approx 60$ in 2D, $N = 216$ in 3D), single-engine and classical, and say nothing about excitons, string theory or the universal-optimality theorem. Written with an AI system; not refereed.

1 What was tested, and why

The suggestions came from a brainstorm that borrowed vocabulary from string theory: "lift to 8D or 24D where the optimum is perfect and project back"; Kaluza–Klein "dimensional hopping"; boundaries as flexible branes. Our own analysis (the project note in the repository) concluded that no theorem transports the optimality of E_8 or the Leech lattice to 3D and that the difficulty of N-body minimisation is the number of local minima, not the dimension. What survives as a testable statement is modest: whether extra coordinates help a local optimiser escape defective minima, whether giving the wall extra freedom reduces frustration at the boundary, and how a

relaxed CM point set compares as a mesh with the standard CVT. All three were registered before the first run (`exploration/exciton/dimlift/PREREG.md`); three amendments (A1: container penalty and failure rule, A2: $N = 216$ for E3, A3: a speed-only vectorisation) are dated.

Scope. Containers, wall nodes (spacing 0.5), random-sequential-addition starts and the defect analysis are those of the Fenris study, imported unchanged; the optimiser is a new numpy L-BFGS code. Kernels: the ungated dipole $2/r - 2/\sqrt{r^2 + d^2}$ ($d = 0.5$) and the gated kernel $D/d = 3$ (Poisson series, tabulated). Shapes: square, hexagon, disc, 2:1 rectangle, L-shape, annulus (gated kernel on square, hexagon, disc only); $N_{\text{free}} \approx 60$; 30 starts each, shared across methods. No claim is made about excitons.

2 E1: dimensional annealing

Method. The same starts are relaxed directly in 2D, or lifted to $2 + k$ dimensions ($k = 1, 2$; extra coordinates $\mathcal{N}(0, 0.1^2)$; pair energy a function of the lifted distance; wall nodes at $w = 0$), with a quadratic confinement $\lambda|w|^2$ stepped through six values and 150 L-BFGS iterations per stage, then a 2D polish. *A point wall leaks into the extra dimensions*: in development runs 15 of 60 nodes left the container, so the lifted problem uses a container$\times\mathbb{R}^k$ penalty and a projection after each stage (amendment A1). A run with any node outside the container at the end is a failed run (counted, energy excluded; the rule is the same for every method and only the lifted ones fail): 7 of 30 (square, $k = 1$) and 8 of 30 (square, $k = 2$) fail, and 1–4 on the 2:1 rectangle and the L-shape; none elsewhere. The equal-compute control is the best of m direct restarts with m the evaluation ratio (6–17).

Result. Lifting costs 6–17 times more evaluations per start (12–30 times in wall-clock). In none of the nine cases is the mean final energy lower than direct by two standard errors, and in no case does lifting beat the equal-compute control (the best-of-m direct mean is lower than the lifted mean in all nine; panel a). The control is not tilted against lifting: every asymmetry favours it (failed lifted runs consume compute but are not penalised; the best-of-m bootstrap is capped at the sample minimum; the lifted failures come from the final polish of the harness and concentrate on shapes with corners). The fraction of starts reaching the best energy found among the fixed-wall methods (direct/lift1/lift2) is: annulus 0.43/0.40/0.53, disc 0.50/0.40/0.33, hexagon 0.03/0.07/0.10, L-shape 0.00/0.07/0.10, 2:1 0.03/0.00/0.03, square 0.23/0.27/0.17; gated: disc 0.43/0.37/0.27, hexagon 0.07/0.03/0.00, square 0.27/0.17/0.13. With 30 starts these differences are within sampling noise and have no consistent sign (the registered "fraction not lower" conjunct holds in three of nine cases for lift1 and in four of nine for lift2); for gated hexagon, lift2 is significantly worse than direct in a paired comparison ($t \approx -3$). The number of distinct minima (1e-6 relative) is similar across methods (e.g. square 13/12/13, L-shape 20/23/15). The registered gate (dipole kernel: lower mean by more than 2 s.e. on at least four of six shapes, fraction not lower, and surviving the control) is **not met**.

3 E2: sliding walls

Method. Non-corner wall nodes slide along their ring (arclength parameter) with the wall–wall repulsion included, same starts; energies are total energies including the wall–wall term. The sliding problem is a superset of the fixed-wall one, so its minimum energy is lower *by construction*; the informative quantity is the change of the defect count.

Result. The mean change in the number of non-6-coordinated free nodes (slide minus fixed; $\pm$ one standard error) is: square -2.2 ± 1.5, L-shape -1.2 ± 0.9, annulus -0.7 ± 0.4, 2:1 rectangle -0.4 ± 0.4, hexagon $+0.1 \pm 0.2$, disc $+1.7 \pm 0.6$ (dipole kernel); gated: square -0.9 ± 1.0, hexagon 0.0 ± 0.3, disc 0.0 ± 0.7. The registered gate (decrease of at least two on at least three of six shapes) is **not met**: it is met on one (square, -2.2 ± 1.5: the mean is 0.15 s.e. from the threshold

and 1.5 s.e. from zero, so not itself significant), and the disc gets *worse* by about three standard errors. The mean total energy change (slide minus fixed) is negative in seven of nine cases (largest: L-shape -0.087, gated square -0.079) but *positive* on the disc ($+0.010$) and the annulus ($+0.0025$): the sliding problem contains the fixed one, so these are local minima, not global ones. The net topological charge Q changes in 0–40% of starts depending on the shape, against the registered expectation that it would not; we did not test why (the Gauss–Bonnet identity fixes Q given the boundary coordination, which sliding changes, but this is an interpretation).

4 E3: a CM-relaxed periodic point set as a 3D mesh

Method. $N = 216$ points in the periodic unit cube (30 starts) are relaxed (a) with the screened repulsion e^{-kr}/r, $kL = 6$ (minimum image, shifted cutoff at $L/2$), or (b) by 40 Lloyd/CVT iterations with 1.5×10^5 Monte-Carlo samples; Poisson points are the reference. Delaunay tetrahedra of the $3 \times 3 \times 3$ tiling restricted to the central cell are scored by the radius ratio $3r_{\mathrm{in}}/R$ and the minimum dihedral angle.

Result. Both relaxations are far better than Poisson (median 5th-percentile radius ratio 0.20 for Poisson, 0.63 for CM, 0.68 for CVT; fraction of tetrahedra with radius ratio < 0.3: 12% for Poisson, 0.5% for both relaxations). CM is below CVT in the 5th-percentile radius ratio by 0.045 ± 0.007 (paired mean $\pm$ s.e., from the per-seed records) and beats it in 3 of 30 paired starts; the gate (24 of 30) is **not met**. The picture is not one-sided: CM is *better* on the worst-case metrics (median minimum radius ratio 0.112 against 0.086; median minimum dihedral angle 5.9° against 4.4°) while CVT is better on the 5th-percentile radius ratio and dihedral angle (36.8° against 35.0°); the registered gate concerned only the former. The kernel parameter $kL = 6$ was fixed in advance and not tuned; a different screening length may change the margin, which was not tested.

5 What this does and does not show

- The lifted problem is more expensive and no better at equal compute for these small, strongly confined instances. This does not exclude a benefit for other problem classes (the rank-relaxation literature is about different landscapes) or much larger N.
- Sliding walls help one shape marginally and hurt another significantly; no general statement is supported. No buckling into a third dimension was tested.
- CM relaxation is a good point-set regulariser but not better than CVT as measured; nothing is shown about conforming meshes of CAD domains, size grading or boundary recovery.
- None of this involves the lattice-optimality theorem, whose planar statement remains unrefereed and AI-generated.

Process and deviations. The development runs (square seeds 0–5, E3 seed 0) are inside the registered sets and amendment A1 tuned the lifted-run parameters on them. The CVT used 1.5×10^5 Monte-Carlo samples per iteration, not the registered 2×10^5 (not amended; disclosed in the erratum of the pre-registration). Records are in `exploration/exciton/dimlift/results/`; a first set of 76 disc runs produced with a slow nearest-edge search was discarded and the disc rerun from scratch after the speed-only amendment A3. The change is numerically equivalent, not identical: on the square it is bit-identical (18 runs rechecked), on the disc 'direct' and 'slide' are bit-identical but 15 of 37 lifted runs end in a different local minimum (up to 0.12 in energy), so the discarded 76 runs are a usable replicate. The square and hexagon runs predate A3 and are kept. The numbers in this paper are generated by `analyse.py` (`summary.json`); the verifier notes of papers II–III and the independent review of this paper are in `docs/designs/EXCITON_FLUID_VERIFIER_NOTES_*.md`.

References

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